

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 18)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

A nuclear fuel reprocessing plant discharges small amounts of radioactive material in liquid effluent to a coastal sea.

0 1

1

Explain why it cannot be assumed that the discharge will be diluted to insignificant levels.

[2 marks]

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0 1

2

Give **four** factors that would be included in a Critical Pathway Analysis of this discharge.

[4 marks]

1

2

3

4

0 1

3

Explain how the results of the analysis reduce the cost of environmental monitoring.

[2 marks]

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0 1

4

Explain why Critical Group Monitoring measures exposure rather than damage to health.

[2 marks]

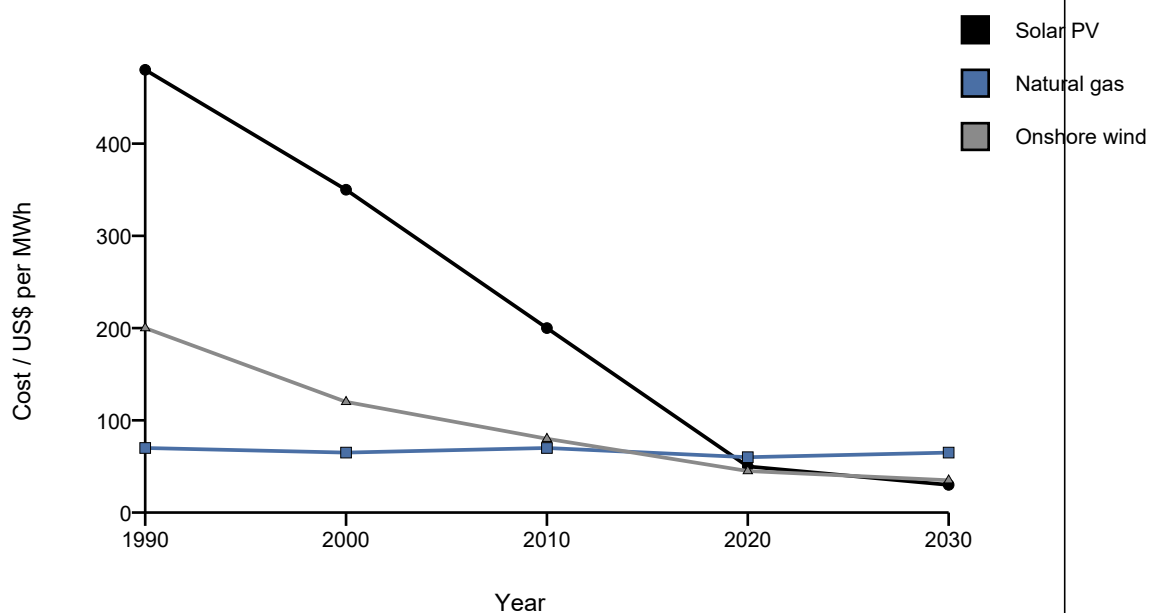
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10

0 2

Figure 1 shows the estimated cost of generating electricity from three sources between 1990 and 2020, and projections to 2030.

Figure 1



0 2

. 1

Describe the trends shown in Figure 1.

[3 marks]

0 2

. 2

Explain why new energy technologies are expensive when first introduced and why their costs then fall.

[3 marks]

0 2 . 3

Explain why a renewable energy project may still be at a financial disadvantage compared with a gas-fired power station of the same total cost.

[2 marks]

0 2 . 4

Explain why the price paid for fossil fuel energy does not reflect its full cost.

[2 marks]**10**

0 3

Table 1 shows some techniques used to locate mineral deposits.

0 3 . 1

Complete Table 1 by adding the name of the technique or a description of how it works.

[4 marks]**Table 1**

Technique	How it works
Gravimetry	
	Detects rocks that are more magnetic, such as magnetite iron ore or ores of tungsten and cobalt
Seismic survey	
	Measures how difficult it is for electricity to pass through rocks; sedimentary rocks with high water content have lower values than igneous rocks

03 . 2

Explain why trial drilling is usually carried out only after remote sensing surveys.

[3 marks]

03 . 3

Explain how infrared spectroscopy from satellites can help identify mineral deposits.

[2 marks]

03 . 4

Give **one** advantage of portable field analysis equipment such as X-ray fluorescence.

[1 mark]

10

0 4 . 1 Explain how ionising radiation damages living cells.

[3 marks]

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0 4 . 2 Distinguish between acute and chronic effects of radiation exposure.

[2 marks]

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0 4 . 3 Distinguish between **exposure** and **contamination**.

[2 marks]

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0 4 . 4 A worker receives an absorbed dose of 0.002 Gy from alpha radiation.
Calculate the effective dose in sieverts.

[2 marks]

Answer

Sv

0 4 . 5 State what is measured by the becquerel.

[1 mark]

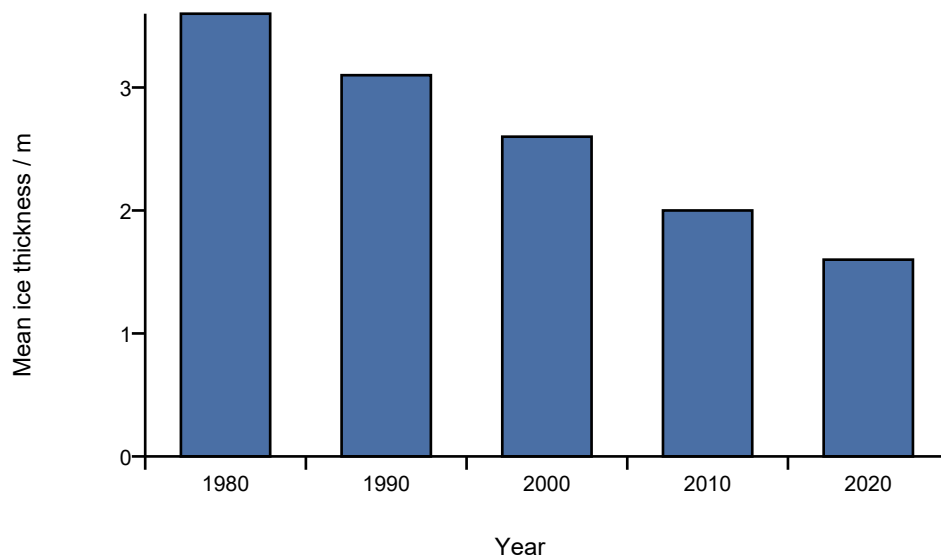
10

0 5 Table 1 shows the area of Arctic sea ice in September (the annual minimum) in selected years. Figure 1 shows the mean thickness of Arctic sea ice in the same years.

Table 1

Year	1980	1990	2000	2010	2020
September sea ice area / million km ²	7.7	6.2	6.3	4.9	3.9

Figure 1



0 5 . 1 Calculate the percentage decrease in September sea ice area between 1980 and 2020.

[2 marks]

Answer

%

0 5 . 2

Explain why the loss of sea ice does **not** directly raise sea level, but the loss of land ice does.

[2 marks]

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0 5 . 3

Explain how the changes shown in Table 1 and Figure 1 could lead to further warming.

[3 marks]

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0 5 . 4

Explain **two** ways in which rising temperatures cause sea level to rise.

[2 marks]

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0 5 . 5

Explain why the break-up of an ice shelf can increase the rate at which land ice flows into the sea.

[1 mark]

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10

06 . 1

Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content

☐

B High clay content

☐

C Crumb peds

☐

D High pH

☐

06 . 2

Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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06 . 3

Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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5

07

Table 1 shows global reactive nitrogen fixed each year by natural processes and by human activities.

Table 1

Source	Nitrogen fixed / million tonnes per year
Biological fixation (natural ecosystems)	110
Lightning	5
Haber process (fertiliser manufacture)	120
Cultivation of legume crops	60
Combustion (NO _x)	30

07 . 1

Use Table 1 to calculate the percentage of total nitrogen fixation that is caused by human activities.

[2 marks]

Answer %

07 . 2

Describe the Haber process and explain why it is unsustainable compared with biological nitrogen fixation.

[3 marks]

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07 . 3

Describe **three** consequences of the increased movement of nitrogen compounds caused by human activities.

[3 marks]

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3

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07 . 4

Explain how composting or anaerobic digestion of biological wastes produces a better fertiliser than raw waste.

[2 marks]

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10

08 . 1

Explain how each of the following human activities alters processes in the hydrological cycle.

[6 marks]

Deforestation

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Urban development

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Agriculture

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08 . 2

Describe **two** ways in which global climate change alters the hydrological cycle.

[2 marks]

1

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2

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08 . 3

Explain why the impact of industrial and domestic water use on the hydrological cycle is usually local.

[2 marks]

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10

09

A student measured the wind speed at 10 sites inside a woodland and at 10 sites in a nearby clearing.

Table 1

Location	Mean wind speed / m s^{-1}	Standard deviation / m s^{-1}
Woodland	1.2	0.4
Clearing	3.5	1.1

09

1

Name the statistical test the student should use to find out whether the difference between the means is significant. Give **two** reasons for your choice.

[3 marks]

09

2

Explain what the student should do if it is not possible to show that the data are normally distributed.

[1 mark]

09

3

The test gave $p = 0.02$. Explain what this means.

[2 marks]

09

4

Explain how the standard deviations in Table 1 support the conclusion that the means are different.

[2 marks]

09 . 5

Explain why a statistically significant result does not prove that trees cause the lower wind speed.

[2 marks]

10

10 . 1

Explain how the properties of smoke and of the gases involved in acid rain lead to their environmental effects, and evaluate the effectiveness of the methods used to control them.

[9 marks]

[illegible]

10 . 2

State **one** effect of tropospheric ozone on plants.

[1 mark]

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10

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Write an essay on one of the following topics.

1	1	.	1
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Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

1 1 . 2

Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a full page of a worksheet designed for handwriting practice. It features 20 evenly spaced, horizontal dotted lines across the entire page, providing a guide for letter height and placement. The background is plain white, and there are no margins or additional markings.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS